

What an 8-Bit Frame Can and Cannot Say About the Scene Behind It

Bounds for single-image inverse tone mapping, and a gate that reaches one of them

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Abstract

We treat single-image inverse tone mapping as a question about information rather than about model capacity. A compact residual model on an analytic inverse tone map is scored against that same analytic map on 429 held-out frames, in PU21-PSNR and in ColorVideoDP (CVVDP) JOD. On degraded input, the condition in which real footage arrives, it gains 1.43 dB and 0.44 JOD and wins 348 of 429 frames. On clean, well-graded input it loses 3.0 dB of PU21-PSNR for a perceptually invisible -0.046 JOD, the two metrics disagreeing sharply about the practical significance of the same difference, which is itself one of our results. The regression is not diffuse: it is carried by low-dynamic-range frames, and the model has no representation of how much headroom a frame has, so it cannot decide to do nothing.

We then measure what repairing that is worth and whether it can be repaired. An oracle per-frame scale on the residual is worth 5.84 dB on clean and 0.29 dB on degraded input at the same time, and no constant achieves it. But a linear readout of the features such a gate would see explains 3% of that scale’s variance, and 79% of the variance lies within a condition rather than between conditions, so even a perfect clean-versus-degraded classifier is capped at $R^2 = 0.213$. The residual variance asks whether a clipped region was a 200 cd/m² lamp or a 20,000 cd/m² sun. Capacity, corpus and objective were each varied by a factor of four to six and none of them moved the result. We report this as an empirical identifiability bound on the representations we tested, not as an information-theoretic one.

That bound is a property of the question. Posed as a continuous per-frame scale it is unreachable; posed as a binary decision on the one axis that is detectable, it is reachable. A 21,121-parameter gate on the shadow arm of the reconstruction converts the clean regression into a gain of $+3.41 \pm 0.32$ dB and $+0.135 \pm 0.023$ JOD over the shipped model, for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD given up on degraded input, across three seeds.

1 Introduction

An 8-bit delivery frame is the end of a lossy chain. Scene radiance spanning perhaps twenty stops was metered, exposed, graded through a tone curve chosen by a colourist, clipped at both ends and quantised to 256 levels per channel. Single-image inverse tone mapping asks for the radiance back. The task is attractive because the input is abundant and the output is what every high dynamic range (HDR) pipeline needs, and it is hard for a reason that is easy to state and easy to forget: the forward map is not injective, and no amount of model does anything about that.

The literature has largely treated the problem as one of architecture and training. We treat it as one of information, and the distinction turns out to be load-bearing. This paper reports a

compact model, the conditions under which it helps and hurts, a measurement of the ceiling on adapting it per frame, and a component that reaches a particular part of that ceiling by asking a different question of the same input.

Position of this work. Our model is deliberately small (1,196,197 parameters) and is a residual on an analytic inverse tone map rather than a direct predictor. That choice is what makes the paper’s central question askable at every point in training and evaluation: *does the network beat doing nothing?* A direct predictor has no “nothing” to be compared against, and in practice a great deal of what a learned inverse tone mapper appears to contribute is recoverable analytically. Every number in this paper is a difference against that analytic baseline on identical inputs.

Contributions.

1. **A conditional characterisation of when learned inverse tone mapping helps.** On degraded input the model gains 1.43 dB PU21-PSNR and 0.44 JOD over the analytic baseline, on 81% of held-out frames. On clean input it loses 3.0 dB PU21-PSNR for -0.046 JOD. Reporting either condition alone misrepresents the method (Section 6).
2. **A characterised disagreement between the two standard HDR metrics.** On the same frames and the same models, PU21-PSNR reports a substantial fidelity penalty where CVVDP reports a perceptually negligible difference, and the two agree decisively when the difference is large. The two are not commensurable as numbers, which is the point: what differs is the practical significance each assigns to a small reconstruction difference on well-graded input. We localise that divergence, show it is only partly carried by the error tail, and show it reappearing inside a checkpoint-selection rule that touches neither metric (Sections 6, 7 and 10).
3. **A localisation of the failure to a specific mechanism.** The clean regression is not invented highlights, which was our own first two hypotheses. It is the shadow arm of the residual gate firing on scenes that need no reconstruction, and disabling that arm inverts the regression (Section 7).
4. **A measured bound on per-frame adaptation.** An oracle per-frame residual scale is worth 5.84 dB on clean and 0.29 dB on degraded input simultaneously, and is poorly identifiable from the input we have: a linear readout of the features the gate sees explains 3% of its variance, and 79% of the variance lies within a condition rather than between, which caps *any* condition-detection architecture at $R^2 = 0.213$ regardless of how it is built. Capacity, corpus and objective were each varied by a factor of four to six without moving it. We are careful to claim an empirical identifiability result and not an information-theoretic one; Section 8 separates what the measurements establish from what they suggest.
5. **A component that reaches the reachable part of that bound.** Posed as a binary decision on the detectable axis rather than as a continuous scale, the problem is tractable. A 21,121-parameter gate on the shadow arm delivers $+3.41 \pm 0.32$ dB and $+0.135 \pm 0.023$ JOD on clean input for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD on degraded input across three seeds, and is the only configuration we scored that is positive on all four measures (Section 9).
6. **Three defects in our own evaluation harness, reported rather than quietly repaired.** Checkpoint selection on the maximum of a noisy series, an evaluation that silently read the alphabetical front of its split, and a viewer that presented every frame upside down beneath

tests that compared only float buffers. Each corrupted a result we believed, and each is available to any comparable pipeline (Section 10).

Scope. This paper reports a direct SDR-pixel to scene-linear-radiance image model. It does not report the DRE-transformer and cross-attention architecture described in an earlier manuscript from this group: that design was never trained, and the results attributed to it were aspirational. Both are withdrawn. Appendix D states what was withdrawn and why, so that a reader who has seen the earlier document can reconcile the two. What follows is the system that exists, measured with the metrics that exist.

2 Related work

A note on how these references are used. Every reference below was checked against the publisher or author page for authors, title, venue and year, and the one-line method descriptions are at the level the titles and abstracts support. No number in this paper is attributed to any of them. We compare against our own analytic baseline and against ExpandNet run from its authors’ released weights (Section 6.3); Section 11 states which comparisons are still missing. The failure mode this caution replaces, placeholder tables presented as measurements, is documented in Appendix D.

Single-image inverse tone mapping. The dominant framing learns a mapping from an 8-bit frame to a higher-range one, typically with an encoder-decoder and a loss in a perceptual or log domain, and typically evaluated on synthetically clipped data. Eilertsen et al. [1] predict a log-domain reconstruction of the saturated regions with a U-Net and blend it back through a saturation mask; Marnerides et al. [10] combine three branches at different receptive fields; Endo et al. [2] instead synthesise a bracketed exposure stack from the single input and merge it; Liu et al. [7] decompose the task into learned inverses of the individual camera-pipeline stages. Our architecture is deliberately smaller than this line of work and is a residual on an analytic inverse tone map rather than a direct predictor, which is what makes “does the network beat doing nothing?” a question we can ask at every step of training.

Highlight and clipped-region reconstruction. A related family treats the problem as inpainting the saturated regions specifically, rather than remapping the whole frame. Santos et al. [11] make the clipped area explicit by masking it out of the network’s own features and adding a perceptual loss, and the masked blend of Eilertsen et al. [1] serves the same end. Our highlight and shadow masks share that purpose but are predicted jointly with the residual and gated by a luminance prior rather than by a detected saturation mask. Section 7 is a direct criticism of that choice: our own failures are not in the clipped highlights at all.

Evaluation of HDR reconstruction. PU21 [8] and ColorVideoVDP [9] are the two public measuring sticks we use, the first an encoding that makes existing metrics such as PSNR applicable to absolute-luminance HDR, the second a calibrated visible-difference predictor reporting in JOD. Section 6 reports a case where they diverge sharply in the practical significance they assign to the same difference on the same frames, and Section 7 shows the PU21 penalty is only partly tail-carried. We are not aware of that disagreement having been characterised for this task, and consider it a contribution independent of the model.

Training data for HDR. The scarcity of paired SDR/HDR footage shapes every result here. Public HDR video with cinematic production values remains small enough to enumerate: the HDM-HDR-2014 set of Froehlich et al. [3] contributes 9 scenes and 11,007 frames to our corpus, and the widely used HDR still sets [6] are bracketed-exposure captures rather than graded footage. The remainder of our corpus is proprietary. Section 5.1 documents a distribution shift between our own training and held-out splits that we did not design and that a reader should weigh.

What we compare against. ExpandNet [10] is scored on our split in Section 6.3, using the authors’ own model and released weights and their own pre- and post-processing. Our benchmark harness scores any third-party output against the same reference, so the remaining comparisons are a matter of running published code rather than of building anything; Section 11 says which are outstanding.

3 Problem formulation

3.1 Units and storage

Absolute units are not a presentation detail in this problem; conflating two conventions is a multi-stop error that produces entirely plausible numbers. We therefore fix them once. HDR targets are stored in a `log2_extended` encoding covering 0.005 cd/m^2 to $1,000,000 \text{ cd/m}^2$ at 2,377 codes per stop. *Network units* are cd/m^2 divided by 10,000, so $L = 1.0$ denotes $10,000 \text{ cd/m}^2$ and diffuse white, taken at 203 cd/m^2 after ITU-R BT.2408 [5], is $L = 0.0203$. The network clamps its output at $L_{\max} = 4.0$, that is $40,000 \text{ cd/m}^2$. Storage elsewhere in the pipeline uses the delivery convention in which 1.0 is diffuse white; the two differ by a factor of 49.26 and are never mixed within a computation.

3.2 The forward map, and where it is not invertible

Let $L \in \mathbb{R}_{\geq 0}^{H \times W \times 3}$ be scene radiance in network units. An 8-bit delivery frame $s \in [0, 1]^{H \times W \times 3}$ is produced by

$$s = Q_8\left(\text{clip}_{[0,1]}(T(L))\right), \quad (1)$$

where T is a display-referred transform, in general an unknown composition of exposure, a creative grade and a tone curve; `clip` is the delivery range limit; and Q_8 is quantisation to 256 levels per channel.

Two distinct failures of injectivity follow, and the distinction matters for everything after Section 6. Where $T(L) \geq 1$ the frame is *clipped*: all radiances above the clipping point share one code. Where $T(L)$ falls below the first few codes the frame is *crushed*: a range of low radiances shares a small number of codes, and quantisation noise there is large relative to the signal. Inverting Eq. (1) exactly is possible only on the interior, where T is monotone and the quantisation step is small relative to the local slope.

3.3 Censored observations

Graded HDR sources carry a delivery ceiling c : $4,000 \text{ cd/m}^2$ for HDM-HDR-2014, approximately $1,000 \text{ cd/m}^2$ for Rec.2100 PQ 1K, $10,000 \text{ cd/m}^2$ for Chimera. A target pixel sitting on that ceiling states $L \geq c$, not $L = c$. Treating it as an equality trains the model to cap its own highlights, which is the specific failure the ceiling-aware loss exists to prevent, and 78% of the image corpus and 84% of the video corpus are graded to such a ceiling.

We therefore treat those pixels as right-censored observations. Writing $\ell(x) = \log(1 + \kappa x)$ for the log-radiance encoding with $\kappa = 16$, the per-pixel error is

$$e = \begin{cases} [\ell(L) - \ell(\hat{L})]_+ + [\ell(\hat{L}) - \ell(\min(2^3 c, 10^6))]_+ & \text{if } L \geq c(1 - 10^{-3}), \\ |\ell(\hat{L}) - \ell(L)| & \text{otherwise,} \end{cases} \quad (2)$$

with $[\cdot]_+ = \max(0, \cdot)$. Under-prediction of a censored pixel is charged; over-prediction is free for `CENSORED_HEADROOM_STOPS` = 3 stops above the ceiling and charged beyond. Without that second term a purely one-sided loss has no upper anchor and the cheapest behaviour for a censored pixel is to run to $L_{\max}$. Three stops takes a 4,000 cd/m² graded clip to 32,000 cd/m², which covers real specular highlights without licensing invention. The 10^{-3} relative slack absorbs a `uint16` log round trip in the stored target.

Censoring is a small effect on the evaluation corpus even though it shapes training: the mean censored fraction is 0.0003.

3.4 What “reconstruction” can and cannot mean here

Given Eq. (1), a method that maps s to $\hat{L}$ is not recovering L on the clipped set; it is estimating a plausible value consistent with the observation $L \geq c$. On the interior it is inverting a transform, which is a different and far better-posed operation. The results in Sections 6 and 8 are best read with that split in mind: the analytic baseline is strong precisely because most pixels of most frames are interior, and the learned component earns its place where they are not.

4 Method

4.1 Analytic baseline

The baseline is an analytic inverse of the transform used to render the corpus SDR side,

$$b(s) = \text{ACES}^{-1}(\text{sRGB}^{-1}(s)) \cdot \frac{2 \cdot 203}{10,000}, \quad (3)$$

in network units. The factor 203/10,000 places diffuse white, and the factor 2 is the -1 EV the corpus preparation applies before the ACES curve. Equation (3) is therefore *exact* for corpus frames and approximate for SDR produced by any other pipeline, which is a property of the baseline and not of the model; the learned component is what absorbs other camera and tone curves. Every comparison in this paper is against Eq. (3) on identical inputs, so the exposure convention cancels.

4.2 Composite

A compact U-Net f_θ (1,196,197 parameters, 4.6 MiB in float32) takes the SDR frame and its baseline and predicts three fields: a log-domain residual r , a highlight mask m_h and a shadow mask m_s . Let $y = \text{luma}(s)$ be Rec.709 luma of the SDR frame. The residual is applied through a fixed per-pixel prior

$$p_h = \sigma(24(y - 0.82)), \quad p_s = \sigma(24(0.10 - y)), \quad g = \max(p_h, w \cdot p_s), \quad (4)$$

where $w \in [0, 1]$ is a per-frame weight on the shadow arm, fixed at 1 for the shipped model and predicted by the gate of Section 9. The composite is

$$\ell(\hat{L}) = \text{clip}_{[0, \ell(L_{\max})]}(\ell(b(s)) + r \cdot g), \quad (5)$$

$$\hat{L} = \ell^{-1}(\ell(\hat{L})), \quad \ell(x) = \log(1 + \kappa x), \quad \kappa = 16. \quad (6)$$

Outside the recovery region the baseline is restored explicitly,

$$\hat{L} \leftarrow b(s) + \max(m_h, m_s) \cdot (\hat{L} - b(s)), \quad (7)$$

so that the physically grounded analytic inverse survives through ordinary midtones and learned capacity is spent where Eq. (1) was not invertible.

Two properties of Eq. (4) carry the rest of the paper. First, the prior is a function of *one pixel’s brightness and nothing else*: it has no access to the frame, and therefore no representation of how much headroom the scene has. Section 7 shows this is the mechanism behind the clean regression. Second, the two arms are separable, and $w = 1$ and $w = 0$ reproduce exactly the two configurations that Section 7.3 measures, which is what makes the gate of Section 9 an interpolation between measured endpoints rather than a new model.

4.3 Training objective

Write e for the censored per-pixel error of Eq. (2), h and s for the ground-truth highlight and shadow masks, and $\rho = \max(h, s)$ for the recovery region. The objective is

$$\begin{aligned} \mathcal{L} = \underbrace{\bar{e}}_{\log L_1} &+ 0.50 \mathcal{L}_{\text{hi}} + 0.20 \mathcal{L}_{\text{sh}} + 0.10 \mathcal{L}_{\text{chroma}} + \lambda_{\text{sc}} \mathcal{L}_{\text{shadow-chroma}} + \lambda_{\text{ss}} \mathcal{L}_{\text{shadow-smooth}} \\ &+ 0.10 \mathcal{L}_{\text{edge}} + 0.05 \mathcal{L}_{\text{mask}} + 0.05 \mathcal{L}_{\text{outside}} + 0.10 \mathcal{L}_{\text{resid-outside}}, \end{aligned} \quad (8)$$

where $\mathcal{L}_{\text{hi}}$ and $\mathcal{L}_{\text{sh}}$ are e averaged over the highlight and shadow masks, $\mathcal{L}_{\text{chroma}}$ is an L_1 on normalised chromaticity, $\mathcal{L}_{\text{shadow-chroma}}$ is an L_1 on opponent channels within the shadow mask (chromaticity ratios are poorly conditioned near black, where they force legitimate shadows toward neutral grey), $\mathcal{L}_{\text{shadow-smooth}}$ penalises residual gradients inside the shadow mask, $\mathcal{L}_{\text{edge}}$ is a gradient term on log radiance, $\mathcal{L}_{\text{mask}}$ is binary cross-entropy on the two predicted masks, and the last two terms charge error and residual magnitude *outside* ρ , which is what holds Eq. (7) honest during training. The two λ weights are schedule parameters; the remainder are fixed.

5 Experimental setup

5.1 Corpus and splits

The corpus holds 27,678 training, 435 validation and 429 test records; validation and test are 97 scenes each at 1280×720 . Sampling is scene-balanced with 40% video mass. The only publicly redistributable component is the HDM-HDR-2014 set of Froehlich et al. [3], 9 scenes and 11,007 frames at a median 5.99 stops of measured dynamic range; the remainder is proprietary footage, which is why Appendix B ships the manifests and the scoring harness rather than the corpus. The manifest is pinned by SHA-256 in every checkpoint’s configuration, so a run can be tied to the exact record set it saw.

5.2 An unintended distribution shift

The held-out splits are not drawn from the training distribution (Table 1). The band in which the model fails, below 400 cd/m^2 , is 45.2% of the test split and 27.2% of what the sampler draws. We report this because it is a confound in our own results; Section 8.2 argues that it is not the binding constraint, and Section 11 records that we have not retrained under a matched sampler to prove that.

Table 1: The held-out splits are lower in dynamic range than the training distribution the sampler draws from. Peak is the reference peak luminance of the frame.

	median peak	< 1,000 cd/m ²	< 400 cd/m ²	video share
train, as sampled	1,713 cd/m ²	42.2%	27.2%	40.0%
validation	548 cd/m ²	69.0%	38.2%	33.8%
test	546 cd/m ²	58.0%	45.2%	32.9%

5.3 Training configuration

Table 2 gives the shipped configuration (v5) and the capacity ablation (v6), which differ only in width. Both were trained on a single NVIDIA RTX 4080 SUPER.

Table 2: Training configuration. v6 differs from v5 only in base width, and is the capacity ablation of Section 6.4.

	v5 (shipped)	v6 (capacity)
base channels	32	64
parameters	1,196,197	4,770,117
crop	384	384
batch $\times$ grad-accum	2×2	same
optimiser	AdamW, lr 2×10^{-4} , wd 10^{-4} , cosine to 5%	same
gradient clip	1.0	same
degradation probability	0.65	same
scene-balanced sampling	yes, 40% video mass	same
steps	100,000	100,000
evaluation	every 1,000 steps, 256 held-out crops, both conditions	same
selection	trailing median of <code>composite_gain</code> (Section 10.1)	same
seed	20,260,715	20,260,715

5.4 Conditions, metrics and protocol

Conditions. CLEAN is the held-out frame as prepared, a well-graded plate. HARD applies a seeded camera and codec degradation to the same frame: an unknown tone curve, 4:2:0 chroma subsampling, banding and JPEG. HARD is the deployment condition; CLEAN is the condition in which the analytic baseline is already strong.

Reference. Both conditions are scored against the same *unclamped* reference. Clamping the reference at the network’s own ceiling would score the model against a ground truth cropped to its own limits, which flatters it exactly where it is weakest.

Metrics. PU21-PSNR [8] in decibels and ColorVideoVDP [9] in JOD, both at `-nits-scale 203`. A JOD is a just-noticeable difference unit; differences well below 0.1 JOD are not expected to be visible. Reporting the two together is not redundancy: Section 6.2 shows they can put a substantial

penalty and a perceptually negligible one on the same comparison. The two are in different units and are not directly comparable as numbers; what is comparable is whether each calls a difference significant.

Reported quantity. Unless stated otherwise, every number is a *paired difference* against the analytic baseline of Eq. (3) on the same frame under the same condition, so corpus difficulty cancels. “Frames won” counts frames on which that paired difference is positive, out of 429.

Statistical protocol. The headline comparisons are over all 429 held-out frames, paired, and we report the mean and median of the per-frame difference rather than the difference of means. The oracle and ceiling analyses of Section 8 use one frame per scene (27 and 51 frames respectively) for cost reasons, and Section 11 quantifies the bias that introduces. Where a result rests on a margin comparable to run-to-run variation, we retrain and report mean $\pm$ standard deviation across seeds: Section 9.3 does this for the gate with $n = 3$. Three runs support the claim that a sign is stable; they do not support a confidence interval, and we do not present one.

6 Results

6.1 The model against its own analytic baseline

Table 3 gives absolute scores for every configuration on the 429 held-out frames, and Table 4 the paired per-frame differences for the shipped model.

Table 3: Absolute scores on 429 held-out frames. The shipped model (v5) is the best configuration on degraded input and the worst on clean; only the gated configuration of Section 9 is at or above the analytic baseline on both.

Condition	Method	PU21-PSNR (dB)	CVVDP (JOD)
clean	analytic baseline	45.99	9.448
clean	v5 (32 ch, step 81,000)	42.99	9.402
clean	v6 (64 ch)	43.24	9.452
clean	v5 + shadow gate	46.06	9.561
hard	analytic baseline	25.92	7.362
hard	v5	27.34	7.805
hard	v6	26.88	7.706
hard	v5 + shadow gate	27.15	7.751

Table 4: Paired per-frame difference, v5 against the analytic baseline, 429 frames. The clean PU21 loss of 3.0 dB corresponds to -0.046 JOD, far below a just-noticeable difference.

	mean Δ PU21	median Δ	frames won	mean Δ JOD	median Δ JOD
hard	+1.427 dB	+0.609	348 / 429 (81%)	+0.443	+0.134
clean	-2.999 dB	-2.131	115 / 429 (27%)	-0.046	-0.040

The metrics disagree in magnitude on clean input

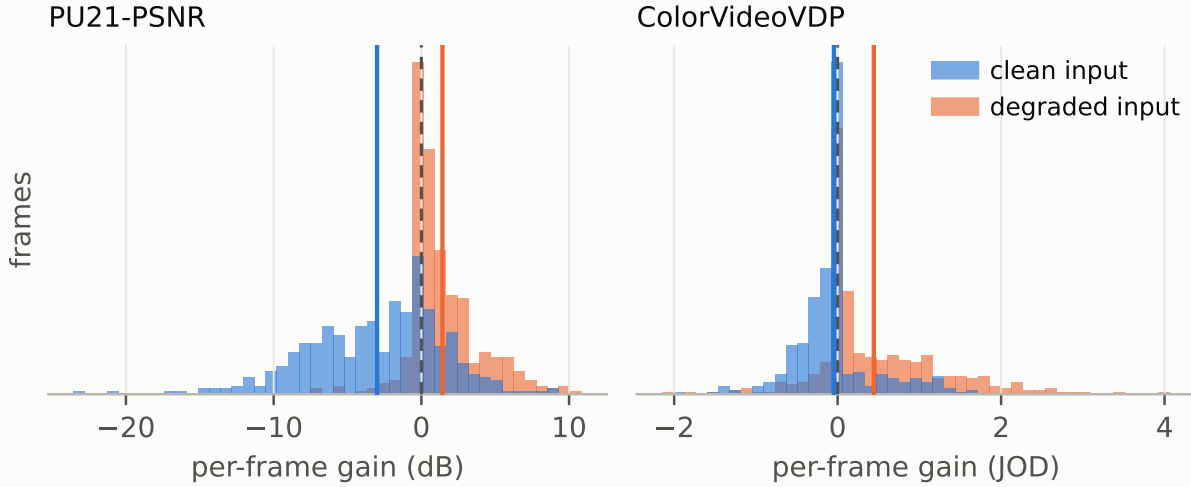


Figure 1: Per-frame gain over the analytic baseline, 429 held-out frames; vertical rules mark the means. The clean distribution is broad and left-shifted in PU21-PSNR and collapses onto zero in CVVDP. The same frames, the same models, two metrics, one calling the difference substantial and the other calling it invisible.

6.2 Fidelity penalty, perceptual indifference

A 3.0 dB PU21-PSNR loss corresponds to -0.046 JOD (Fig. 1), far below a just-noticeable difference. What the model adds to well-graded input is highlight energy that PU21-PSNR punishes and that no viewer sees. The clean PU21 row should never be reported without the JOD beside it. Section 6.3 shows the converse case, in which the two agree decisively, and Section 10.1 shows the disagreement reappearing inside a selection rule that touches neither metric.

6.3 A published method on the same split

Every number above is against our own analytic baseline, which has been this paper’s largest gap since the first draft. It is now closed for one method. ExpandNet [10] was run on the same 429 frames using the authors’ released weights rather than a reimplementaion, with their pre-processing and post-processing unchanged, on the exact 8-bit frames our model was given (Table 5).

Table 5: Clean condition, 429 frames. ExpandNet is -18.56 dB and -2.029 JOD against our deployed configuration, winning 1 of 429 frames on PU21 and 16 on CVVDP.

Method (clean, 429 frames)	PU21 (dB)	CVVDP (JOD)
RUDRA + gate, as deployed	46.06	9.561
RUDRA + gate, exposure-aligned	46.61	9.558
analytic inverse-ACES baseline	45.99	9.448
v5 backbone, no gate	42.99	9.402
ExpandNet	27.50	7.532

The exposure alignment, and what it is worth to each side. ExpandNet ends in a sigmoid and its released post-processing min-max normalises each frame to $[0, 1]$, so it predicts *relative* radiance with no absolute anchor. Scoring that directly against a reference in cd/m^2 would measure its exposure guess. We therefore fit one global scalar per frame on the pixels the SDR input neither clipped nor crushed. That is a free parameter and it is not free to grant: ExpandNet’s fitted scale has a $35.5\times$ spread across frames (p10 0.49, p90 17.45), where ours is $1.0\times$ (p10 0.9993, p90 1.018) because our model predicts absolute cd/m^2 . Put through the same fit our model gains 0.55 dB and *loses* 0.0025 JOD, so the row we report for ourselves is the unaligned one.

The alignment is not what costs ExpandNet its score. Given the *best possible* per-frame exposure, chosen to maximise its own PU21, it recovers 0.39 dB.

Three caveats, because a gap this size invites the question. ExpandNet is run outside its training domain: our SDR frames come from an ACES approximation at a -1 EV offset, not the curve its corpus used. Our reference is unclamped and reaches roughly a million cd/m^2 , a range no method trained on display-referred targets was built for. And ExpandNet compresses dynamic range by a median factor of $2.3\times$ against the reference where ours is within $1.32\times$, which contributes to but does not explain 18 dB. The row should be read as *this method, unmodified, on this corpus*, not as a ranking of the field. Santos et al. [11], Eilertsen et al. [1] and Liu et al. [7] have public code and have not been run.

What this settles about Section 6.2. The two metrics part company on our model against its own baseline and agree decisively here: 18.6 dB and 2.03 JOD say the same thing, and 2 JOD is two just-noticeable differences. The divergence is a property of *small* differences on well-graded input, not a defect in either instrument. When the difference is real, both see it.

6.4 Capacity is not the constraint

v6 quadruples width to 4,770,117 parameters. It is better on clean (0.25 dB, +0.05 JOD) and worse on hard (-0.46 dB , -0.10 JOD) than v5: movement in both directions, and smaller than the spread between conditions. Section 8.3 places this beside the corpus and objective ablations.

7 Localising the clean regression

7.1 It is a function of scene headroom

The clean regression is concentrated, not diffuse. Splitting clean frames by the headroom the ground truth actually has separates them cleanly (Table 6), and the trend is monotone across the full range (Fig. 2). The correlation between $\log_2$ of the reference peak and per-frame gain is +0.46.

Table 6: Clean frames split by the headroom of the reference. The model’s error is a function of the scene.

clean frames	mean Δ vs baseline	median reference peak
60 worst	-10.76 dB	238 cd/m^2
60 best	3.41 dB	19,590 cd/m^2

The mechanism is Eq. (4). The per-pixel luminance prior cannot distinguish a 238 cd/m^2 studio interior from a 20,000 cd/m^2 sunset, because it never sees the frame.

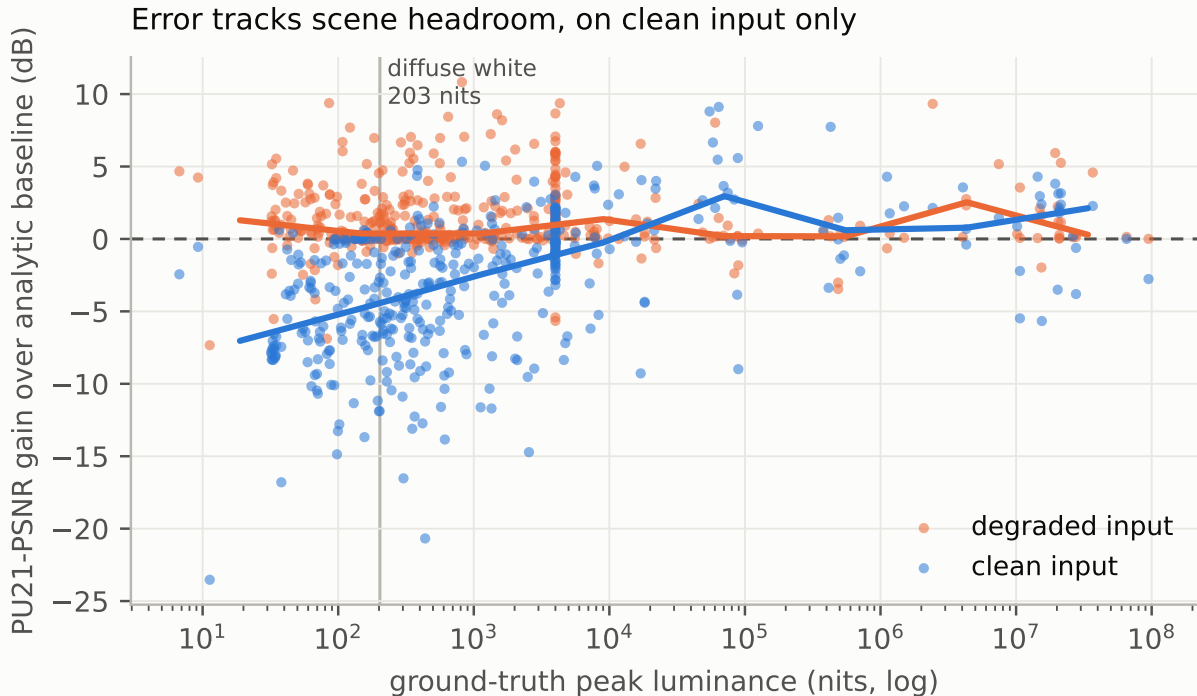


Figure 2: Per-frame gain against reference headroom, log x -axis, with binned medians. On clean input the trend rises through zero at roughly $10,000 \text{ cd/m}^2$; on degraded input it is flat and positive at every headroom.

7.2 The error is in the shadows, and only partly in the tail

On 51 clean held-out frames we computed the per-pixel PU21 squared error for both the model and the baseline, then recomputed each PSNR with the worst pixels progressively discarded (Table 7). If the penalty were carried by a few bad pixels the gap would close.

Table 7: Trimmed PU21-PSNR on clean frames below 400 cd/m^2 , $n = 25$. Discarding a tenth of every frame removes 2.3 dB of the 5.55 dB gap and leaves 3.2 dB: the penalty is partly tail-carried and mostly broad.

discard worst	baseline	RUDRA	gap
—	54.40	48.84	− 5.55 dB
0.1%	54.70	49.40	−5.30
1%	55.36	51.14	−4.23
10%	57.58	54.34	− 3.23 dB

The worst 1% of pixels carry 32% of the model’s total squared error, but the bulk of the distribution is genuinely worse too. At those worst pixels the *true* luminance has median 4 cd/m^2 against 21 cd/m^2 frame-wide: they are darker than typical, not brighter, and only 46% of them are over-predicted. On frames at or above 400 cd/m^2 the model and the baseline are level (-0.03 dB) and the worst pixels move to the bright end instead (median 560 cd/m^2 against 57 cd/m^2 frame-wide).

This relocates the failure. Equation (4) has two arms, and on a low-dynamic-range scene the shadow arm fires on content that needs no reconstruction at all. The headroom correlation of $+0.46$ is real, but the mechanism behind it is the *shadow* path, not invented highlights. That is a specific

and testable target, and one that the single global scale of Section 8 could not have isolated.

7.3 The shadow arm, ablated

The hypothesis is directly testable: disable the shadow prior, setting $w = 0$ in Eq. (4), and rescore the same 429 frames (Table 8).

Table 8: Gain over the analytic baseline with the shadow arm on and off, 429 frames. Turning it off does not shrink the clean regression, it inverts it.

configuration	clean		hard	
	dB	JOD	dB	JOD
shadow arm on (shipped)	−3.00	−0.046	+1.43	+0.443
shadow arm off	+0.51	−0.017	+0.33	+0.134

The hypothesis is confirmed, and the result is a trade rather than a free win. Turning the shadow arm off moves clean by 3.51 dB, so the model now beats the analytic baseline on clean input by 0.51 dB, and costs 1.10 dB on degraded input. The shadow path is therefore the whole of the clean regression and simultaneously carries most of the degraded-input gain, which is what one would expect: crushed shadows are exactly what a bad tone curve produces and exactly what a well-graded frame does not have.

This is the observation that makes Section 9 possible, and Section 8 is what makes it necessary to state the problem that way.

8 A bound on per-frame adaptation

8.1 What an oracle gate is worth

Give the composite a single global scale α on the residual, so that $g \leftarrow \alpha g$ in Eq. (5), and let an oracle choose it per frame (Table 9, 27 held-out scenes).

Table 9: PU21 gain over the analytic baseline as a function of a global residual scale α , 27 held-out scenes. Per-frame adaptation beats the shipped configuration on both conditions at once, which is not a trade-off at all.

α	clean	hard
1.0, as shipped	−4.15	+1.08
0.125, best single constant	+0.83	+0.17
per-frame oracle	+1.69	+1.37

The two conditions want opposite settings. Clean improves monotonically as α falls to ≈ 0.125 , hard as it rises to ≈ 1.1 , and no constant serves both: the constant that repairs clean discards 85% of the degraded-input gain. Per-frame adaptation beats the shipped configuration by 5.84 dB on clean and 0.29 dB on hard simultaneously.

The oracle’s choice is also legible (Table 10). On clean input it is near-binary in headroom; on degraded input it wants full strength regardless, because degradation destroys information the baseline cannot recover whatever the scene’s range. A gate would need to see headroom *and* condition together.

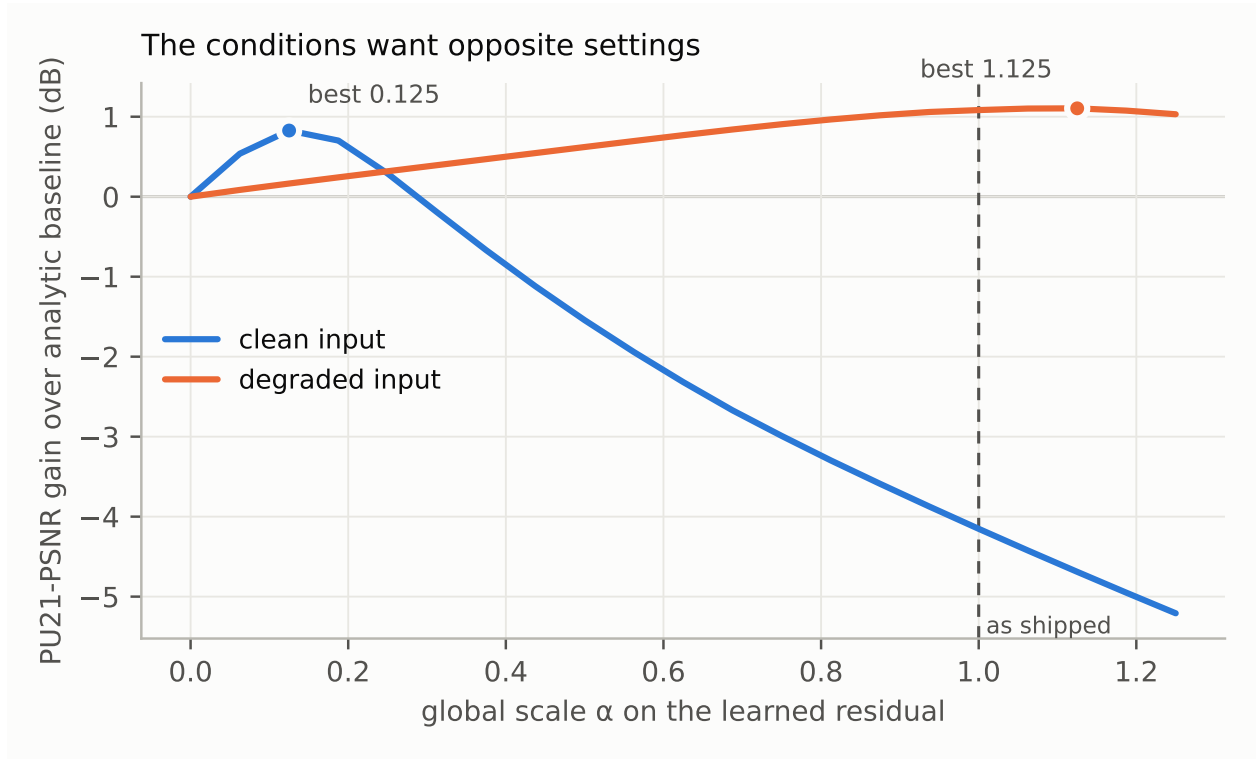


Figure 3: Mean gain as a function of the global residual scale α . The curves cross: every α that helps one condition hurts the other, and the shipped $\alpha = 1$ is near-optimal for degraded input and near-worst for clean.

Table 10: Median oracle α , split by reference peak.

	peak < 1,000 cd/m ²	≥ 1,000 cd/m ²
clean	0.125	0.969
hard	1.250	1.094

8.2 The ceiling

We built that gate, a pooled-feature head predicting α (21,121 parameters), and trained it three times: twice through the reconstruction loss, 16,000 steps in total, and once by direct supervision on the oracle. All three produced a near-constant output. The trained head emits $\alpha \in [1.0125, 1.0802]$ with correlation -0.037 against $\log_2$ of the reference peak, where the oracle wanted 0.125.

Rather than attempt a fourth variant we measured the ceiling directly. On 51 held-out frames, each clean and degraded, giving 102 samples, we fit a linear readout on exactly the features the head receives, cross-validated *by frame* so that a clean/degraded pair can never straddle the split (Table 11).

Table 11: Predicting the oracle α from what the gate can see. Frame-grouped cross-validation, 102 samples.

knowing	MAE on oracle α	R^2
nothing (predict the mean)	0.501	0.000
the features, via a linear readout	0.487	+0.031
the condition, <i>perfectly</i>	0.409	+0.213
the oracle itself	0.000	1.000

Two numbers close the question. The features explain 3% of the target’s variance. And 79% of that variance is *within* a condition and only 21% *between* (clean α : mean 0.436, sd 0.471; degraded: mean 0.925, sd 0.469), so a perfect clean-versus-degraded classifier, which is the most any condition-detection architecture could buy, is capped at $R^2 = 0.213$. Since 55% of frames want α at one end of the grid, the target is close to a binary decision per frame.

The remaining 79% asks whether *this* frame’s clipped region was a 200 cd/m² lamp or a 20,000 cd/m² sun, which is a question about scene radiance that the delivery frame may not settle.

What this does and does not establish. It is worth separating the two, because the temptation to state the stronger version is exactly the error Section 10.4 records us making twice already.

Established. The variance decomposition is a property of the oracle target and the condition labels, not of any model: 79% of the variance in α lies within a condition, so no architecture that works by detecting the condition, however good, can explain more than $R^2 = 0.213$. That ceiling is real and holds for any method of that shape.

Measured, but of one representation. The 3% figure is what a *linear* readout of the pooled features *this particular head sees* recovers, on 102 samples from 51 frames. It bounds that readout and that representation. It does not bound what some other representation might hold: the signal could be nonlinear in these features, or spatial where these are pooled, or semantic, or present in the frame and absent from anything we extracted.

Not established. That the information is absent from the 8-bit frame. We varied capacity, corpus and objective by a factor of four to six (Section 8.3) and nothing moved, which is evidence against a trivial architecture fix and is not a proof of unidentifiability. A formal statement would need a bound we have not derived. What we claim is the empirical version: *the oracle scale is poorly identifiable from the single-frame representations we tested*, and the 5.84 dB it is worth stayed out of reach of every attempt we made to predict it.

What is detectable. The one axis that carries signal is the condition itself. A frame-grouped cross-validated classifier on the same features reaches **75.5%** accuracy at separating clean from

degraded, against 3% of the variance explained for the continuous target. That asymmetry is the whole opening, and Section 9 walks through it.

8.3 Three levers, one bound

The bound is not an artefact of any one design choice. Each of the three levers usually reached for was varied by a factor of four to six (Table 12).

Table 12: None of the three standard levers moves the bound.

lever varied	change	effect
capacity	1.20M $\rightarrow$ 4.77M parameters	± 0.5 dB, sign varies (Section 6.4)
corpus	6 $\times$ footage	0.03 dB
objective	censored loss; oracle supervision	no movement in α

9 The shadow gate: reaching the reachable part of the bound

9.1 Reformulating the question

Section 8 asks a hard question: predict a continuous per-frame scale, of which only 21% of the variance is explained by the clean/degraded distinction and only 3% by the available features. Section 7.3 makes a different question available. The shadow arm is a *binary* decision, and its correct setting is *exactly* the clean-versus-degraded axis, which is the part that is partially detectable at 75.5%.

The consequence is arithmetic. At that measured accuracy, a switch on the shadow arm alone is worth 2.65 dB of clean for 0.27 dB of hard (Table 13). That was the prediction we set out to test.

Table 13: Predicted value of a switch on the shadow arm at the measured 75.5% classifier accuracy, against the shipped configuration.

	clean	hard
shipped	-3.00 dB	1.43 dB
75.5%-accurate arm switch	-0.35 dB	1.16 dB

9.2 The gate, trained

SHADOWGATE is 21,121 parameters on the frozen v5 backbone, predicting the single weight w of Eq. (4) per frame. By construction $w = 1$ reproduces the shipped configuration exactly and $w = 0$ reproduces the ablated configuration exactly, so it interpolates between the two measured endpoints of Table 8 and nothing else. It is supervised by binary cross-entropy against the degradation label, and unlike the residual scale of Section 8 that target needs no oracle: at training time we know whether we degraded the frame.

The gate beat the prediction on both axes: predicted -0.35 dB clean and 1.16 dB hard, delivered $+0.07$ and $+1.24$ on the first seed (Table 14) and $+0.41 \pm 0.33$ and $+1.12 \pm 0.15$ across three (Section 9.3). Against the shipped model that is $+3.41 \pm 0.32$ dB and $+0.135 \pm 0.023$ JOD on clean

Table 14: Gain over the analytic baseline, 429 held-out frames. The gated configuration is the only row that is positive on all four measures.

method	clean dB	clean JOD	hard dB	hard JOD
v5, as shipped	−3.00	−0.046	+1.43	+0.443
v5, shadow arm off	+0.51	−0.017	+0.33	+0.134
v6, 4× capacity	−2.75	+0.004	+0.96	+0.344
v5 + shadow gate	+0.07	+0.113	+1.24	+0.389

for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD on hard: it converts the clean regression into a win while keeping most of the degraded-input gain.

Two properties deserve stating precisely. First, it is the only configuration we scored that is positive on all four measures; every other row buys one column with another. Second, its clean CVVDP of 9.561 exceeds *both ends of the ablation it interpolates* (9.402 with the arm on, 9.431 with it off) and every fixed alternative including the 4× capacity model. A hard switch could not do that. The gate is emitting intermediate weights and finding per-frame settings that neither extreme reaches, which is more than the binary framing that motivated it predicted.

9.3 Three seeds

The 0.07 dB clean margin above is small enough that one run proves nothing, so we trained two further gates on the same backbone, changing only the seed, and scored them on the same 429 frames (Table 15).

Table 15: Three gate seeds, gain over the analytic baseline. CVVDP and frames won are on clean, out of 429; the shipped v5 wins 115 for comparison.

seed	clean		hard		clean	clean
	dB	JOD	dB	JOD	CVVDP	frames won
20,260,901	+0.07	+0.113	+1.24	+0.389	9.561	251 (59%)
2	+0.71	+0.068	+0.96	+0.308	9.516	315 (73%)
3	+0.45	+0.089	+1.18	+0.358	9.537	280 (65%)
mean ± sd	+0.41	+0.090	+1.12	+0.352		
	±0.33	±0.023	±0.15	±0.041		

All three seeds are positive on all four measures, and all three reach a clean CVVDP above every fixed alternative we scored (best of those: 9.452). The result is a property of the method rather than of a seed.

It is worth recording which way the single-run report erred. Seed 20,260,901 is the *worst* of the three on clean PU21 (+0.07 against a mean of +0.41) and the *best* on clean CVVDP (+0.113 against +0.090). Reporting it alone understated the decibel result by a factor of six and overstated the JOD result by a quarter. The honest headline is the mean with its spread, and that is what the abstract carries.

9.4 An observation about the selection criterion

Each gate’s checkpoint is chosen by a trailing median over degradation-classification accuracy on validation: 69.5%, 64.1% and 64.8% for seeds 20,260,901, 2 and 3. Ranked by that accuracy, the three runs come out in exactly the order of their clean CVVDP and in exactly the reverse order of their clean PU21 (Table 16).

Table 16: Validation classification accuracy against clean-condition outcome, three seeds. With $n = 3$, a perfect agreement or inversion arises by chance with $p = 1/6$; this is an observation, not a result.

seed	val accuracy	rank	clean JOD	rank	clean dB	rank
20,260,901	69.5%	1	+0.113	1	+0.07	3
3	64.8%	2	+0.089	2	+0.45	2
2	64.1%	3	+0.068	3	+0.71	1

We report it because it is the disagreement of Section 6.2 appearing a third time, after the metrics themselves and after the selection rule of Section 10.1, now in a criterion that touches neither metric. Whatever the gate learns that makes a frame classifiable also makes it look better and measure worse. Three seeds cannot settle that; it is the first thing we would test with thirty.

10 Evaluation methodology: three defects in our own harness

We report these because each silently corrupted a result we believed, and each is a mistake available to any comparable pipeline.

10.1 Selection on a maximum over noise

The shipped checkpoint was chosen by the *maximum* of the composite criterion

$$\text{composite_gain} = \text{hard_gain} + \min(0, \text{clean_gain}) \tag{9}$$

over 102 in-training evaluations. Across that run `clean_gain` had mean -1.43 dB and standard deviation 1.29 dB, with only 10 of 102 evaluations positive; the shipped checkpoint’s $+0.02$ ranked 9th of 102. Its promised hard gain of 1.80 dB measured 1.43 dB on held-out frames, between the evaluation mean ($+1.19$) and its selected maximum, which is exactly where an inflated in-training figure should land. Selection now uses a trailing median over five evaluations; replayed on the same series it selects step 72,000 instead of 81,000.

We then scored step 72,000 on the same 429 frames, and the answer is not the clean vindication we expected. On the criterion the selector optimises it is better: composite gain -1.36 dB against the shipped checkpoint’s -1.57 dB, and clean PU21 improves by 0.55 dB. On CVVDP it is *worse on both conditions*, -0.052 JOD clean and -0.071 JOD hard, so the smoothed rule would have shipped a checkpoint that a perceptual metric likes less. This is the disagreement of Section 6.2 reappearing inside the selection rule itself: median smoothing removes the noise it was designed to remove, and the objective it smooths is still PU21. Smoothing a selector does not fix choosing the wrong quantity to select on. We report the fix and its limit together, because reporting only the first would repeat the error this section is about.

10.2 An evaluation that read the front of the split

`DataLoader(val, shuffle=False)` with `max_batches=N` evaluates the alphabetically *first* N records: 32 records across 11 scenes, every name between `abandoned_factory` and `blau_river`, with 403 records and 87 scenes never measured. Since error correlates with headroom (Section 7.1), that slice reported a clean gain of +0.61 where the benchmark measures -3.0 . Fixed with a seeded permutation applied identically to both conditions.

10.3 A viewer that presented every frame upside down

The default framebuffer places row 0 at the bottom and the display shader sampled its texture coordinate unchanged. No numeric test caught it, because they all compare the float composite, which a presentation flip leaves untouched, and the one test that read the canvas did so through `readPixels`, which returns rows bottom-first and cancelled the flip against itself. Master EXR output was never affected.

10.4 Two claims we corrected against fresh measurement

Section 7.2 has been rewritten twice against new measurements. The first draft claimed the model invents highlights; the second claimed the penalty was tail-carried. Both were wrong, and both were plausible. We record this beside the harness defects because the mechanism is the same: a hypothesis that explains the headline number is not thereby established, and in both cases the measurement that settled it was cheap.

11 Limitations

Held-out size. 429 test frames over 97 scenes. The oracle and ceiling analyses of Section 8 use 27 and 51 frames respectively, one per scene; that sample runs approximately 1.15 dB pessimistic on clean against the full set (-4.15 dB at $\alpha = 1$ where the full 429 read -3.00 dB). Ordering is reliable; absolute values will move.

Distribution shift. The train/test shift of Section 5.2 is a confound in the clean result. Section 8.2 argues it is not the binding constraint, but we have not retrained under a matched sampler to prove it.

The oracle is an upper bound. It reads the ground truth.

Single-image only. The temporal refiner is not evaluated here; its held-out set is 4 validation and 5 test clips of one scene each, too small to report.

The analytic baseline is privileged on clean corpus frames. Equation (3) is the exact inverse of the transform that rendered the corpus SDR side, so on clean frames it is being handed the forward pipeline. It is an unusually strong control, which is what makes the comparison sharp, and it is also unusually strong in a way no baseline would be on footage from elsewhere. A clean-condition regression against it should be read as “worse than knowing the exact forward transform”, not as a general statement about learned inverse tone mapping against whatever baseline a real pipeline would have. The degraded condition does not have this property, since no analytic inverse knows the degradation.

One degradation model, and a gate supervised on it. HARD is our own seeded camera and codec pipeline, not a corpus of real degraded footage. This bears harder on the gate than on the model: the gate is trained by binary cross-entropy against *our* degradation label, so what it learns may be the signature of this generator, JPEG blocking, 4:2:0 chroma, the particular banding and tone-curve family, rather than the property we want, which is that a frame’s shadows need reconstruction. Nothing in our evaluation separates the two, because the test condition is generated the same way as the training condition. The experiment that would separate them is to hold out a degradation family entirely, train on the rest and test on it, and better still to score real degraded footage that our pipeline never touched. Until that is run, the gate’s 75.5% should be read as accuracy on this degradation model.

One published method, not four. ExpandNet is scored on our split (Section 6.3); Santos et al., Eilertsen et al. and Liu et al. have public code and are not. One comparison answers “compared to what?” and does not rank the field, and the caveats of Section 6.3, namely training-domain mismatch, an unclamped reference and a per-frame exposure fit, apply to the one row we have.

No claim here rests on a reimplementing of prior work. Section 2 places the result among families of methods and cites them bibliographically; it does not reproduce their numbers, and we did not run their code.

Three seeds is a spread, not a distribution. The gate is positive on all four measures in all three runs, but $n = 3$ supports “the sign is stable”, not a confidence interval. The clean PU21 spread (+0.07 to +0.71) is wide relative to its mean.

Deployment carries the gate as a scalar only. The interactive viewer receives the predicted shadow weight per frame and multiplies the shadow prior by it in the compositing shader; CPU/GPU parity holds to 8.4×10^{-6} relative error across 11 cases spanning $w = 0$ to $w = 1$. The weight cannot be folded into the residual the way a scalar residual scale can, because it enters inside the max of Eq. (4) and is not linear in the residual.

12 Conclusion

A 1.2M-parameter residual on an analytic inverse tone map is worth 1.43 dB and +0.44 JOD on degraded input, on 81% of held-out frames, and costs 3.0 dB of PU21-PSNR on clean input for a perceptually invisible -0.046 JOD. Its defect is that it does not know when to do nothing.

We measured what repairing that is worth, an oracle per-frame scale being 5.84 dB on clean, and then measured how much of that scale is predictable from what the model can see. A linear readout of the features the gate receives explains 3% of its variance, and 79% of the variance lies within a condition rather than between, so even a perfect clean-versus-degraded classifier is capped at $R^2 = 0.213$. Capacity, corpus and objective were each varied by a factor of four to six; none moved it. The ceiling on condition-detection is a property of the target and holds for any architecture of that shape; the 3% is a statement about one representation and one linear readout, and we do not claim to have shown the information is absent from the frame.

The bound is a property of the question. Asked for a continuous scale, the input cannot answer. Asked a binary question on the axis that *is* detectable, whether this frame arrived clean or degraded, it can. Locating the failure in the shadow arm rather than in the highlights made that decomposition available, and a 21,121-parameter gate on that arm delivers $+3.41 \pm 0.32$ dB and $+0.135 \pm 0.023$ JOD

on clean for 0.30 ± 0.15 dB and 0.091 ± 0.041 JOD on hard across three seeds: the only configuration we scored that is positive on all four measures, in every run, with a clean CVVDP that exceeds both ends of the ablation it interpolates.

The lesson we would carry to another problem is not about tone mapping. Four of the things that cost us most in this work were measurement defects rather than model defects. Three are in Section 10: selection on the maximum of a noisy series, an evaluation that read the alphabetical front of its split, and a viewer that presented every frame upside down beneath tests that only ever compared float buffers. The fourth is in Section 9.3, and is the one we would warn a reader about first: our initial report of the gate quoted a single seed that happened to be the weakest of three on one metric and the strongest on the other. We would not have known without running the other two.

The ceiling we could prove was real, the one we could only measure held against everything we threw at it, and so were the ways we nearly failed to see either.

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A Notation

Table 17: Symbols used throughout.

symbol	meaning
L	scene radiance in network units; $1.0 = 10,000 \text{ cd/m}^2$
s	8-bit SDR frame, $[0, 1]^{H \times W \times 3}$
T	unknown display-referred transform (exposure, grade, tone curve)
c	delivery ceiling of a graded source
$b(s)$	analytic inverse-tone-map baseline, Eq. (3)
r	learned log-domain residual
m_h, m_s	learned highlight and shadow masks
p_h, p_s	fixed per-pixel highlight and shadow priors, Eq. (4)
g	per-pixel residual gate
w	per-frame shadow-arm weight, predicted by the gate of Section 9
α	global scale on the residual, Section 8.1
κ	log encoding scale, $\kappa = 16$
$L_{\max}$	network output ceiling, $4.0 = 40,000 \text{ cd/m}^2$
$\ell(x)$	$\log(1 + \kappa x)$

B Reproducibility

The benchmark is four exports and six scorings, resumable:

```
# the full benchmark
powershell -ExecutionPolicy Bypass -File training\run_bench.ps1

# one more checkpoint against the same reference
powershell -ExecutionPolicy Bypass -File training\score_checkpoint.ps1 `
    -Checkpoint <ckpt> -Name <label>
```

```
# every derived number in the results sections recomputed from those
# results and diffed against the paper; exit status 1 on any mismatch
python training\audit_paper_numbers.py --bench <bench dir>

# the headroom split and its correlation, from the same results joined
# to the corpus manifest's per-frame peak luminance
python training\analyze_headroom.py --bench <bench dir> `
    --manifest <manifest> --check
```

Results land as a results document, per-clip JSON and per-frame CSV. The composite of Eq. (5) exists in three languages, PyTorch, GLSL and NumPy, pinned against each other to 8.4×10^{-6} relative error by a parity test. The test suite covers those, canvas orientation (Section 10.3), PU21 torch-versus-numpy parity, and a smoke test that exercises all 54 controls of the interactive viewer.

The audits above are not decoration. An earlier draft reported 346 of 429 frames won on degraded input where the benchmark says 348; the count had been transcribed by hand, and nothing in the pipeline compared it back to the file it came from. An earlier draft gave v6 4,772,485 parameters where the checkpoint has 4,770,117, for the same reason. Every derived number in Sections 6, 7 and 9 is now recomputed on demand, and both scripts exit non-zero on any drift.

Two analyses remain uncovered by that audit: the trimmed-PSNR table of Section 7.2 and the oracle sweep of Section 8.1. Both need per-pixel statistics over the reference frames rather than the per-frame results the benchmark writes, and we state that rather than leaving a reader to assume otherwise.

Weights, frame pairs and HDR sources are not distributed; the training scripts regenerate them.

C Standards referenced by the pipeline

- SMPTE ST 2084:2014, the PQ electro-optical transfer function of mastering reference displays [13].
- ITU-R BT.2100, image parameter values for high dynamic range television [4].
- ITU-R BT.2408, operational practices in HDR television production, from which diffuse white at 203 cd/m^2 is taken [5].
- SMPTE ST 2065-4:2013, the ACES image container file layout, AP0 chromaticities [12].

D Relation to an earlier manuscript

An earlier manuscript from this group described a DRE-transformer and cross-attention architecture and reported results for it. Two things in that document are withdrawn without qualification.

First, its results tables quoted an “HDR-VDP-3” column that corresponds to no metric this codebase ever computed. The only HDR perceptual metric implemented is ColorVideoVDP, which reports JOD on a roughly 0 to 10 scale. Those tables were placeholders and should never have been presented as measurements.

Second, the architecture those tables described was never trained to completion, so rows implying an evaluated system were aspirational.

The present paper reports a different system: a direct SDR-pixel to scene-linear-radiance image model, measured with the metrics that exist. We record the withdrawal here rather than silently omitting it, because a reader who has seen the earlier document is entitled to know which claims

were retired and why. The caution about citations at the head of Section 2, and the audit scripts of Appendix B, are both direct responses to that failure.